Lean certificates for Kitaev's ε-C^* rigidity beyond finite dimensions

October 18, 2026

Abstract

This note certifies the two positive results of the paper *Kitaev's ε-C^* rigidity beyond finite dimensions*: Theorem A (Kitaev's approximate factorization of an almost-idempotent channel on a Hilbert space of any dimension, when $\widetilde{\Phi}$ has finite rank) and Theorem B (infinitely many finite sectors, including block-diagonal channels). Theorem B is covered by two certificates, one for its first sentence and one for the rest. All three certificates are formalized in Lean 4 with Mathlib and pass an audit consisting of a source scan, a full build, an exact type match, an axiom check and an independent kernel replay. All three are *conditional*: the results of Kitaev's paper that they rely on, and three textbook facts about completely positive maps, appear as named hypotheses of the formal statement. For each certificate we give the formal statement, a dictionary from Lean to mathematics, the assumptions and how faithful they are, a non-vacuity check, a map from the proof to the Lean declarations, and what the formalization found.

1 Summary

Result	Lean theorem	Status	Audit
Theorem A: $\Phi \approx \Delta\Upsilon$ through a C^* algebra, any $\dim \mathcal{H}$, $\widetilde{\Phi}$ of finite rank	`Cert.KitaevChannels.main` (19 modules, about 3,600 lines)	conditional, 6 named assumptions	PASS, 2026-10-16 11:48:58 +0800
Theorem B, first sentence: ℓ^∞-sums of finite-dimensional ε-C^* algebras	`Cert.KitaevSectors.main` (3 modules)	conditional, 1 named assumption	PASS, 2026-10-16 11:49:20 +0800
Theorem B, the rest: extended algebras, and $\Phi = \bigoplus_k \Phi_k(P_k \cdot P_k)$	`Cert.KitaevSectorSums.main` (10 modules, about 1,800 lines)	conditional, the 6 named assumptions of Theorem A	PASS, 2026-10-18 03:12:11 +0800

Toolchain `leanprover/lean4:v4.35.0-rc4`, Mathlib commit `1f414401f69059aa7eead47b53ee40bd38455eeb`.
The Lean sources accompany this note as `lean_sources.zip`, which unpacks to the Lake project `lean/`.

2 How to read a certificate

A reader who wants to rely on a certificate needs to trust three things, and none of them is the proof.

1. *The formal statement* says what the paper says. It is the file `Statement.lean`, which contains definitions only. All three files are reproduced verbatim in the appendices, and each part below explains them in a dictionary.

2. *The assumptions* are the cited results, faithfully stated. They are explicit hypotheses of the proposition `Claim`, never `axiom`s, and a certificate is conditional exactly when such hypotheses are present.

3. *The Lean kernel* is sound.

The audit checks everything else mechanically, in this order:

1. a scan of every source file, with comments and strings stripped, for `sorry`, `admit`, `axiom`, `native_decide` and similar escape hatches, with extra rules for `Statement.lean` (no theorems, no instances, no notation);

2. a full build of every module;

3. a type check that `main` proves exactly `Claim`, via a generated `example : Claim := main`;

4. `#print axioms` for `main` and for every non-vacuity theorem, where only Lean's standard `propext`, `Classical.choice` and `Quot.sound` are allowed;

5. an independent replay of every module by `leanchecker`.

3 Theorem A: approximate factorization on a Hilbert space of any dimension

3.1 Informal statement

There are universal constants $\eta_0, C > 0$ with the following property. Let $\mathcal{H} \neq 0$ be a Hilbert space of any dimension and Φ a UCP map on $\mathbf{B}(\mathcal{H})$ with $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \leq \eta \leq \eta_0$, such that $\widetilde{\Phi}$, defined by Kitaev's (12.2), has finite rank. Then there are a finite-dimensional C^* algebra $\mathcal{B} = \bigoplus_j \mathbf{M}_{d_j}(\mathbb{C})$, an extended $C\eta$-isomorphism $v\colon \mathcal{B} \to \operatorname{Im}\widetilde{\Phi}$ (in particular a linear bijection), and UCP maps $\Delta\colon \mathcal{B} \to \mathbf{B}(\mathcal{H})$, $\Upsilon\colon \mathbf{B}(\mathcal{H}) \to \mathcal{B}$ with $\|\Delta\Upsilon - \Phi\|_{\mathrm{cb}} \leq C\eta$ and $\|\Upsilon_n(\Delta_n(X)\Delta_n(Y)) - XY\| \leq C\eta\|X\|\|Y\|$ for all n and $X, Y \in \mathbf{M}_n(\mathcal{B})$. The proof gives $\eta_0 = \min(\eta_1, \eta_2, \varepsilon_3/(c_1+1), 1/(2\cdot 10^5\Lambda))$ and $C = 10^6\Lambda$ with $\Lambda = C_3c_1 + c_2 + 1$, in terms of the constants of the three cited results of Kitaev.

3.2 Formal statement

The proposition `Claim` of `Statement.lean` (SHA-256 `b6a7086255f9…`; the whole file is Appendix A):

```
def Claim : Prop :=
  CPMapFacts → StinespringThm → UnitaryDesign →
  ∀ (c₁ η₁ c₂ η₂ C₃ ε₃ : ℝ), 0 ≤ c₁ → 0 < η₁ → 0 ≤ c₂ → 0 < η₂ → 0 ≤ C₃ → 0 < ε₃ →
  Kitaev12_1 c₁ η₁ → Kitaev12_6 c₂ η₂ → Kitaev10_5 C₃ ε₃ →
  ∃ (η₀ C : ℝ), 0 < η₀ ∧
  ∀ (H : Type) [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H] [Nontrivial H]
    (Φ : (H →L[ℂ] H) →L[ℂ] (H →L[ℂ] H)) (η : ℝ),
    IsUCP Φ.toLinearMap → CBBound (Φ * Φ - Φ).toLinearMap η → 0 ≤ η → η < η₀ →
    FiniteDimensional ℂ (LinearMap.range (tildePhi Φ).toLinearMap) →
    ∃ (m : ℕ) (d : Fin m → ℕ) (v : BlockAlg d →ₗ[ℂ] (H →L[ℂ] H))
      (Δ : BlockAlg d →ₗ[ℂ] (H →L[ℂ] H)) (Y : (H →L[ℂ] H) →ₗ[ℂ] BlockAlg d),
      IsExtDeltaIsoOnto (C * η) (LinearMap.range (tildePhi Φ).toLinearMap) (choiEffros Φ) v ∧
```

```
IsUCP Δ ∧ IsUCP Y ∧
CBBound (Δ ∘ₗ Y - Φ.toLinearMap) (C * η) ∧
∀ (n : ℕ) (X Y : Mat n (BlockAlg d)),
  ‖(X.map Δ * Y.map Δ).map Y - X * Y‖ ≤ C * η * ‖X‖ * ‖Y‖
```

3.3 Dictionary

Lean	Meaning	Note
`H →L[ℂ] H`	$\mathbf{B}(\mathcal{H})$, bounded operators on a complex Hilbert space of any dimension	Mathlib's C^*-algebra structure; no separability or dimension assumption
`Mat n M`	$\mathbf{M}_n(M)$ with Mathlib's `CStarMatrix` C^* norm and order	the norm of $\mathbf{M}_n(\mathbf{B}(\mathcal{H}))$ is the operator norm on $\mathcal{H}^n$
`BlockAlg d`	$\bigoplus_{j<m} \mathbf{M}_{d_j}(\mathbb{C})$	every finite-dimensional C^* algebra has this form; blocks with $d_j = 0$ are allowed
`CBBound φ c`	$\|\varphi\|_{\mathrm{cb}} \le c$: every amplification `X ↦ X.map φ` has norm at most c	the definition of the cb norm, written as a bound so that no supremum is needed
`IsCP`, `IsUCP`	complete positivity, and in addition $\varphi(1) = 1$	the defining condition of Mathlib's `CompletelyPositiveMap`
`tildePhi Φ`	Kitaev's (12.2), $\frac{1}{2}\big(1 + (2\Phi - 1)(1 - 4(\Phi - \Phi^2))^{-1/2}\big)$, with the binomial series	converges for $\eta < 1/4$; the bound $\eta_0 \le 1/(2 \cdot 10^5)$ keeps it convergent
`choiEffros Φ`	the approximate Choi–Effros product $x \star x' = \widetilde{\Phi}(xx')$, Kitaev's (12.5)	
`ExtEpsCStarSub ε A mul`	extended ε-C^* algebra inside $\mathbf{B}(\mathcal{H})$ (Kitaev §10)	axioms (2.1), (2.2), (2.5) at every matrix level, exact unit, involution; (2.4) and (2.6) are automatic for subspaces of $\mathbf{B}(\mathcal{H})$
`IsExtDeltaIsoOnto δ A mul v`	extended δ-isomorphism onto $(A, \star)$ (Kitaev §10)	injective, range A, †-preserving, $\|v(1) - 1\| \le \delta$, and at every level the product defect $\le \delta\|X\|\|Y\|$ and $(1-\delta)\|X\| \le \|v_n(X)\| \le (1+\delta)\|X\|$
`η < η₀`	the paper's $\eta \le \eta_0$	the same theorem with a smaller η_0
`FiniteDimensional ℂ (range Φ⁾`	$\widetilde{\Phi}$ has finite rank	the hypothesis of Theorem A, verbatim

3.4 Assumptions

Hypothesis	Source	Faithfulness of the formal version
`CPMapFacts`	Paulsen, *Completely Bounded Maps and Operator Algebras*, Prop. 3.6; positive maps are self-adjoint	exact: for CP φ between unital C^* algebras, $\|\varphi\|_{\mathrm{cb}} \le \|\varphi(1)\|$ and $\varphi(x^\dagger) = \varphi(x)^\dagger$ (equality $\|\varphi\|_{\mathrm{cb}} = \|\varphi(1)\|$ holds, only $\le$ is assumed)

Hypothesis	Source	Faithfulness of the formal version
`StinespringThm`	Stinespring 1955; Paulsen Thm. 4.1 with the easy converse	exact for maps into $\mathbf{B}(\mathcal{H})$; the dilation space is required to live in `Type`, which Stinespring's construction satisfies
`UnitaryDesign`	e.g. the Weyl–Heisenberg group	exact: a finite probability distribution on unitaries of $\mathbf{M}_d(\mathbb{C})$ with the twirl identity; trivial for $d = 0$
`Kitaev12_1 c₁ η₁`	Kitaev, Theorem 12.1 (any $\dim \mathcal{H}$, §12.1)	matches; the $O(\eta)$ is made explicit as $c_1\eta$
`Kitaev12_6 c₂ η₂`	Kitaev, (12.6), for $1_{\mathbf{M}_n} \otimes \Phi$	matches, with $O(\eta)$ explicit as $c_2\eta$, at every matrix level
`Kitaev10_5 C₃ ε₃`	Kitaev, Theorem 10.5	matches for extended ε-C^* algebras realized inside some $\mathbf{B}(\mathcal{H})$, with the C^* algebra in block form, which is how Kitaev's §9 constructs it; not formalized, since its proof needs Lefschetz–Hopf, H-space theory and the implicit function theorem

3.5 Non-vacuity

`Sanity.lean` proves `hyps_satisfiable_infinite_dim`. On $\ell^2(\mathbb{N})$, which is nontrivial and not finite-dimensional, the map $\Phi(x) = \langle e_0, xe_0\rangle 1$ is UCP and exactly idempotent, so $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta$ for every $\eta \ge 0$, and $\widetilde{\Phi} = \Phi$ has rank one. So every hypothesis of `Claim` on $(\mathcal{H}, \Phi, \eta)$ holds on an infinite-dimensional Hilbert space. Complete positivity is proved from scratch: $\Phi_n(Y^\dagger Y)$ is a sum of Gram matrices tensored with 1, and each of those is a sum of $Z^\dagger Z$ by Mathlib's `posSemidef_gram` and `posSemidef_iff_eq_sum_vecMulVec`. The conclusion is not degenerate: v is a linear bijection onto $\mathrm{Im}\,\widetilde{\Phi}$, and Δ, Υ are UCP. Nothing here shows that the six assumptions hold jointly for some constants; for Kitaev's three results that is the content of his paper.

3.6 Step map

Step of the proof	Lean declarations (module)	Note
compressions $x \mapsto A^\dagger \pi(x) A'$	`compress_cp`, `compress_cb` (Plumbing)	cb bound $(\|A\| + \|A'\|)^2$ by polarization
normalization $w^{-1/2}$	`exists_normalizer` (Normalize)	continuous functional calculus, $\|h - 1\| \le 2\omega$
$\|\widetilde{\Phi} - \Phi\|_{\mathrm{cb}} \le 12\eta$	`tildePhi_sub_cb` (Tilde)	series summed in the Banach algebra of bounded maps, cb bounds passed through the sum
$\widetilde{\Upsilon} = v^{-1}\widetilde{\Phi}$, Kitaev (12.25)–(12.27)	`upsTilde`, `v_upsTilde`, `upsTilde_v`, `upsTilde_cb` (Objects)	$\|\widetilde{\Upsilon}\|_{\mathrm{cb}} \le (1 + 12\eta)/(1 - \delta)$
diagonal of $\bigoplus_j \mathbf{M}_{d_j}$	`Design.diag`, `cross_vanish`, `diag_block` (Design)	**repaired**: independent signs per block, see Findings
Δ' CP, $\|\Delta' - v\|_{\mathrm{cb}} = O(\eta)$	`DeltaP_cp`, `DeltaP_sub_cb` (Delta)	positivity of $\mathbf{M}_n(\mathcal{B})$ generated by $Y^\dagger Y$
$\Delta = \Delta'(1)^{-1/2}\Delta'\Delta'(1)^{-1/2}$	`exists_Delta` (DeltaUCP)	
(12.34)–(12.36) at every level	`phi_delta_cb`, `mul35`, `mul36` (DeltaProps), `sandwich_cb` (UpsBlock)	(12.36) from Kitaev's (12.6) at level n
Stinespring $\Delta = W^\dagger \rho W$ instead of (12.37)	`StinespringThm`, in `main`	dimension-free

Step of the proof	Lean declarations (module)	Note
Lemma 12.4	`Rop_comm`, `Rop_norm_ge` (Stine, Lemma124), `Rop_norm_le` (UpsBlock)	no tensor factorization $R = 1 \otimes C$ needed
choice of ξ_j	`norm_le_norm_comp_P`, `exists_zeta`, `J_sandwich`, `T_sandwich` (Zeta)	approximate maximizer, since the norm need not be attained
Stinespring $\Phi = V^\dagger \pi V$; L_j, Υ'_j	`compress_L`, `Delta_sum`, `ups_delta_eq`, `block_bound` (UpsBlock)	Kitaev's computation of $\Upsilon'_j(\Delta(Y))$ as two exact identities of linear maps
chain $\Upsilon' \approx \widetilde{\Upsilon}$ and normalization	`upsPrime_close`, `exists_Ups` (Upsilon)	$\mathcal{B} \neq 0$ because $\|v(1) - 1\| \leq \delta < 1$
conclusions (ii), (iii)	`final_bounds` (Final)	
$\eta = 0$	`upsTilde_ucp_exact`, `nonneg_of_unital_isometry` (Final)	not in Kitaev; see Findings
constants	`arith_delta`, `arith_main` (Arith), `main` (Proof)	every $O(\eta)$ an explicit multiple of $\Lambda\eta$

3.7 What was not checked

The six assumptions, in particular Kitaev's Theorems 10.5 and 12.1 and equation (12.6), which carry the deep part of the finite-dimensional theory. Part (i) of Theorem A is Theorem 10.5 applied to $\operatorname{Im}\widetilde{\Phi}$, so the formal content of (i) is that the hypotheses of 10.5 are met. The paper's Lemma 2.1 (finite rank of Φ implies finite rank of $\widetilde{\Phi}$), its remark on the Riesz projection and its Corollary 2.4 on codes are outside the formal claim. The constants are deliberately crude.

3.8 Findings

Kitaev's diagonal. The diagonal of a block algebra in Kitaev's §12.2 (just after (12.29)) is the product of per-block unitary designs, and that product is not a diagonal once there are two or more blocks. For $\mathbb{C} \oplus \mathbb{C}$ with trivial designs it is $1 \otimes 1$. The cross-block terms survive because they carry the first moments $\sum_t p_{jt} U_{jt}$. The formal proof attaches an independent uniform sign to each block, which kills every cross-block term (`cross_vanish`), while the per-block twirl identity gives the diagonal property that the argument uses (`diag_block`). Only Δ' depends on this, and the theorem is unaffected.

Choi representations. Kitaev's proof uses two Choi representations, (12.37) for Δ and $\Phi(x) = V^\dagger(x \otimes 1)V$ for Φ. The second exists only for normal Φ. The formal proof replaces both with Stinespring dilations, and the replacement for $R_j = 1 \otimes C_j$ needs no tensor factorization: commutation of R_j with the matrix units of block j is enough.

The case $\eta = 0$. When the multiplicity space is infinite-dimensional the norm of C_j need not be attained. An approximate maximizer leaves a slack that cannot give the exact factorization the theorem requires at $\eta = 0$. There the formal proof argues separately: $\delta = 0$, v is a unital complete isometry, a unital isometry reflects positivity of self-adjoint elements, and so $\widetilde{\Upsilon} = v^{-1}\widetilde{\Phi}$ is itself UCP. On paper, a simpler route seems to work as well: the per-block factor $\|C_j \xi_j\|^2$ is central in $\mathcal{B}$ and is divided out by the final normalization, so any ξ_j with $\|C_j \xi_j\| \geq 1/2$ suffices. That variant is not formalized.

A technical point. Mathlib synthesizes the complex continuous functional calculus on $\mathbf{M}_n(\bigoplus_j \mathbf{M}_{d_j}(\mathbb{C}))$, but not the real one for self-adjoint elements. The positivity arguments

were therefore stated for an abstract C^* algebra and then applied.

4 Theorem B, first sentence: infinitely many finite sectors

4.1 Informal statement

Let ε_0, C be constants for which Kitaev's Theorem 2.3 holds. Every ℓ^∞-direct sum of finite-dimensional ε-C^* algebras with $0 \le \varepsilon < \varepsilon_0$ and $C\varepsilon < 1$ is $C\varepsilon$-isomorphic to a C^* algebra: there is a linear bijection $v\colon \mathcal{B} \to \bigoplus_i^\infty A_i$ from a C^* algebra $\mathcal{B}$ that commutes with the involution, preserves unit and product up to $C\varepsilon$, and distorts norms by at most $1 \pm C\varepsilon$. With infinitely many nonzero sectors the direct sum is infinite-dimensional.

4.2 Formal statement

The proposition `Claim` of `Statement.lean` (SHA-256 `552f06db05a4…`; the whole file is Appendix B):

```
def Claim : Prop :=
  ∀ (C ε₀ : ℝ), KitaevFiniteDim C ε₀ →
  ∀ (ε : ℝ), 0 ≤ ε → ε < ε₀ → C * ε < 1 →
  ∀ (ι : Type) [Nonempty ι] (A : ι → Type) [∀ i, NormedAddCommGroup (A i)]
    [∀ i, NormedSpace ℂ (A i)] [∀ i, FiniteDimensional ℂ (A i)]
    (S : ∀ i, EpsCStar ε (A i)) (T : EpsCStar ε (lp A ∞)),
    (∀ f g i, T.mul f g i = (S i).mul (f i) (g i)) →
    (∀ f i, T.star f i = (S i).star (f i)) →
    (∀ i, T.one i = (S i).one) →
    ∃ (B : Type) (_ : CStarAlgebra B) (_ : Nontrivial B),
      Nonempty (DeltaIso (C * ε) (ofCStarAlgebra B) T)
```

4.3 Dictionary

Lean	Meaning	Note
`EpsCStar ε A`	Kitaev's Definition 2.1, an ε-C^* structure on a complex normed space	fields are the axioms verbatim: bilinear, not necessarily associative product; conjugate-linear isometric involution with $(XY)^\dagger = Y^\dagger X^\dagger$; unit with $I^\dagger = I$; (2.1), (2.2), (2.5), (2.6) and $\lvert\lVert I\rVert - 1\rvert \le \varepsilon$
`DeltaIso δ S S'`	Kitaev's Definition 2.2, a δ-isomorphism	`toEquiv` is the linear bijection; the other fields are (2.8), (2.9), commutation with $\dagger$ and $(1-\delta)\lVert X\rVert \le \lVert vX\rVert \le (1+\delta)\lVert X\rVert$
`ofCStarAlgebra B`	a C^* algebra's own product, involution and unit, as a 0-C^* structure	
`CStarAlgebra B`, `Nontrivial B`	Mathlib's unital complete complex C^* algebra with $1 \neq 0$	Kitaev assumes $I \neq 0$ throughout
`KitaevFiniteDim C ε₀`	Kitaev's Theorem 2.3 with explicit constants	the assumption
`lp A ∞`	Mathlib's ℓ^∞-direct sum: bounded families, supremum norm	

Lean	Meaning	Note
the three hypotheses on `T`	"T is the direct-sum ε-C^* structure"	stated as formulas, so the statement file contains no construction; `Sanity.lean` shows such T exist
`C * ε < 1`	ε small enough that a $C\varepsilon$-isomorphism has a bounded inverse	equivalent to the threshold $\min(\varepsilon_0, 1/C)$; see Findings

4.4 Assumption, non-vacuity and step map

The single assumption `KitaevFiniteDim C ε₀` is Kitaev's Theorem 2.3: every finite-dimensional ε-C^* algebra with $0 \le \varepsilon < \varepsilon_0$ admits a $C\varepsilon$-isomorphism from some nonzero C^* algebra. It matches the theorem, with the $O(\varepsilon)$ and "ε sufficiently small" made explicit and quantified over, and with the algebras restricted to universe 0. Each of these differences makes the assumption no stronger than Kitaev's theorem.

`Sanity.lean` shows that the hypotheses on the sectors and on T hold for every $\varepsilon \ge 0$ with countably many sectors (all equal to $\mathbb{C}$), that this direct sum is not finite-dimensional, and that at $\varepsilon = 0$ the conclusion of the assumption is a theorem: the transcribed axioms at $\varepsilon = 0$ rebuild every field of Mathlib's `CStarAlgebra`, which checks the transcription of Definitions 2.1–2.2.

The proof applies the assumption to each sector (`step1`; Kitaev's definition requires completeness, which holds automatically in finite dimension), equips ℓ^∞ of the resulting algebras with Mathlib's C^* structure (`lpCStarAlgebra`), shows that the coordinatewise map is a linear bijection using $C\varepsilon < 1$ (`step3_equiv`), and passes every δ-inequality to suprema (`step4_deltaIso`).

4.5 What was not checked, and findings

Kitaev's Theorem 2.3 itself, the ultraproduct remark and the open general case are outside the formal claim. The extended version and the second sentence of Theorem B are certified in the next section.

The proof is correct as written, and formalizing it made four quiet assumptions explicit. The condition $C\varepsilon < 1$ is needed, since across infinitely many sectors the inverses must be bounded uniformly. The index set must be nonempty, since an empty sum violates $|\|I\| - 1| \le \varepsilon$. No sign hypothesis on C is needed, since it follows from the existence of any δ-isomorphism. Kitaev's remark that an ε-C^* algebra with $\varepsilon = 0$ is the same as a C^* algebra holds formally.

5 Theorem B, the rest: extended algebras and block-diagonal channels

The previous section certifies the first sentence of Theorem B for ε-C^* algebras in the sense of Kitaev's Definition 2.1. This section certifies everything else in Theorem B: the first sentence for *extended* ε-C^* algebras (Kitaev §10), and the second sentence, about block-diagonal channels.

5.1 Informal statement

Let $\varepsilon_0, C_{\mathrm{K}}$ be the constants of Kitaev's Theorem 2.3 (or 10.5). Every ℓ^∞-direct sum of finite-dimensional (extended) ε-C^* algebras with $\varepsilon < \varepsilon_0$ and $C_{\mathrm{K}}\varepsilon < 1$ is (extended) $C_{\mathrm{K}}\varepsilon$-isomorphic to a C^* algebra. Consequently, if $\mathcal{H} = \bigoplus_k \mathcal{H}_k$ and $\Phi(X) = \bigoplus_k \Phi_k(P_k X P_k)$, where each Φ_k is UCP on $\mathbf{B}(\mathcal{H}_k)$ with $\|\Phi_k^2 - \Phi_k\|_{\mathrm{cb}} \le \eta \le \eta_0$ and $\widetilde{\Phi}_k$ of finite rank, and $C\eta < 1$, with η_0, C the constants of

Theorem A, then $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta$, and (i) and (ii) of Theorem A hold with the infinite-dimensional algebra $\mathcal{B} = \bigoplus_k^\infty \mathcal{B}_k$.

In Lean, C_{K} and ε_0 are the constants `C₃` and `ε₃` of `Kitaev10_5`. The condition $C\eta < 1$ was added to the paper because of this formalization (see Findings).

5.2 Formal statement

The proposition `Claim` of `Statement.lean` (SHA-256 `64072867c25a…`; the whole file, with the definitions `IsBlockSumMap`, `IsBlockSumAlg` and `IsExtDeltaIsoOntoGen`, is Appendix C):

```
def ClaimExt : Prop :=
  ∀ (C₃ ε₃ : ℝ), Kitaev10_5 C₃ ε₃ →
  ∀ (ε : ℝ), 0 ≤ ε → ε < ε₃ → C₃ * ε < 1 →
  ∀ (ι : Type) [DecidableEq ι] [Nonempty ι] (H : ι → Type) [∀ k, NormedAddCommGroup (H k)]
    [∀ k, InnerProductSpace ℂ (H k)] [∀ k, CompleteSpace (H k)] [∀ k, Nontrivial (H k)]
    (Ak : ∀ k, Submodule ℂ (H k →L[ℂ] H k)) [∀ k, FiniteDimensional ℂ (Ak k)]
    (mulk : ∀ k, (H k →L[ℂ] H k) →ₗ[ℂ] (H k →L[ℂ] H k) →ₗ[ℂ] (H k →L[ℂ] H k))
    (A : Submodule ℂ (lp H 2 →L[ℂ] lp H 2))
    (mul : (lp H 2 →L[ℂ] lp H 2) →ₗ[ℂ] (lp H 2 →L[ℂ] lp H 2) →ₗ[ℂ] (lp H 2 →L[ℂ] lp H 2)),
    (∀ k, ExtEpsCStarSub ε (Ak k) (mulk k)) → IsBlockSumAlg A mul Ak mulk →
    ExtEpsCStarSub ε A mul ∧
    ∃ (B : Type) (_ : CStarAlgebra B) (_ : PartialOrder B) (_ : StarOrderedRing B)
      (v : B →ₗ[ℂ] (lp H 2 →L[ℂ] lp H 2)), IsExtDeltaIsoOntoGen (C₃ * ε) A mul v

def ClaimChannel : Prop :=
  CPMapFacts → StinespringThm → UnitaryDesign →
  ∀ (c₁ η₁ c₂ η₂ C₃ ε₃ : ℝ), 0 ≤ c₁ → 0 < η₁ → 0 ≤ c₂ → 0 < η₂ → 0 ≤ C₃ → 0 < ε₃ →
  Kitaev12_1 c₁ η₁ → Kitaev12_6 c₂ η₂ → Kitaev10_5 C₃ ε₃ →
  ∃ (η₀ C : ℝ), 0 < η₀ ∧
  ∀ (ι : Type) [DecidableEq ι] [Nonempty ι] (H : ι → Type) [∀ k, NormedAddCommGroup (H k)]
    [∀ k, InnerProductSpace ℂ (H k)] [∀ k, CompleteSpace (H k)] [∀ k, Nontrivial (H k)]
    (Φk : ∀ k, (H k →L[ℂ] H k) →L[ℂ] (H k →L[ℂ] H k))
    (Φ : (lp H 2 →L[ℂ] lp H 2) →L[ℂ] (lp H 2 →L[ℂ] lp H 2)) (η : ℝ),
    (∀ k, IsUCP (Φk k).toLinearMap) → (∀ k, CBBound (Φk k * Φk k - Φk k).toLinearMap η) →
    0 ≤ η → η < η₀ →
    (∀ k, FiniteDimensional ℂ (LinearMap.range (tildePhi (Φk k)).toLinearMap)) →
    IsBlockSumMap Φ Φk →
    IsUCP Φ.toLinearMap ∧ CBBound (Φ * Φ - Φ).toLinearMap η ∧
    ∃ (B : Type) (_ : CStarAlgebra B) (_ : PartialOrder B) (_ : StarOrderedRing B)
      (v Δ : B →ₗ[ℂ] (lp H 2 →L[ℂ] lp H 2)) (Y : (lp H 2 →L[ℂ] lp H 2) →ₗ[ℂ] B),
      IsExtDeltaIsoOntoGen (C * η) (LinearMap.range (tildePhi Φ).toLinearMap) (choiEffros Φ) v ∧
      IsUCP Δ ∧ IsUCP Y ∧
      CBBound (Δ ∘ₗ Y - Φ.toLinearMap) (C * η) ∧
      ∀ (n : ℕ) (X Y : Mat n B),
        ‖(X.map Δ * Y.map Δ).map Y - X * Y‖ ≤ C * η * ‖X‖ * ‖Y‖

def Claim : Prop := ClaimExt ∧ ClaimChannel
```

Every name not defined in this file (`ExtEpsCStarSub`, `CBBound`, `IsUCP`, `tildePhi`, `choiEffros`, `Mat` and the six assumptions) is the one from Theorem A's statement file, unchanged.

5.3 Dictionary

Lean	Meaning	Note
`lp H 2`	the Hilbert direct sum $\bigoplus_k \mathcal{H}_k$	Mathlib's ℓ^2-sum; any index set, any dimensions

Lean	Meaning	Note
`blk k l x`	the block $P_k x P_l$, as an operator $\mathcal{H}_l \to \mathcal{H}_k$	defined as `prj k ∘L x ∘L inj l`; an operator on $\bigoplus_k \mathcal{H}_k$ is determined by its blocks
`IsBlockSumMap Φ Φk`	$\Phi(X) = \bigoplus_k \Phi_k(P_k X P_k)$	stated by blocks: the diagonal blocks of $\Phi(X)$ are $\Phi_k(P_k X P_k)$ and the off-diagonal blocks vanish
`IsBlockSumAlg A mul Ak mulk`	$(\mathcal{A}, \star) = \bigoplus_k^{\infty}(\mathcal{A}_k, \star_k)$, realized block-diagonally on $\bigoplus_k \mathcal{H}_k$	$\mathcal{A}$ is the set of block-diagonal operators with k-th block in $\mathcal{A}_k$, and $\star$ acts block by block; boundedness is automatic. Matrix norms are those of $\mathbf{M}_n(\mathbf{B}(\bigoplus_k \mathcal{H}_k))$
`ExtEpsCStarSub ε A mul`	extended ε-C^* algebra inside $\mathbf{B}(\mathcal{H})$	as for Theorem A
`IsExtDeltaIsoOntoGen δ A mul v`	extended δ-isomorphism from a C^* algebra $\mathcal{B}$ onto $(\mathcal{A}, \star)$	the conditions of Theorem A's `IsExtDeltaIsoOnto`, with an arbitrary unital C^* algebra in place of $\bigoplus_j \mathbf{M}_{d_j}(\mathbb{C})$; $\mathcal{B}$ carries Mathlib's C^*-order, which `IsUCP` needs
`C₃ * ε < 1`	$C_{\mathrm{K}}\varepsilon < 1$	needed for the inverse family to be uniformly bounded
`η < η₀, ∃ η₀ C`	$\eta \le \eta_0$ and $C\eta < 1$, with constants that do not depend on the family	η_0 and C are chosen before the index set, so they are uniform over all families; the formal η_0 builds in $C\eta < 1$
`[Nonempty ι], [Nontrivial (H k)]`	at least one sector, and no zero sectors	Kitaev's standing assumption $I \neq 0$

5.4 Assumptions and non-vacuity

The assumptions are the six of Theorem A, with the same definitions, so their sources and faithfulness are as in the section on Theorem A. Here `Kitaev10_5` gives the sector isomorphisms of the first sentence, the textbook facts give complete positivity and cb-contractivity of $x \mapsto P_k x P_k$, and `Kitaev12_1` is used once more in every sector, to show that $\widetilde{\Phi}_k$ fixes $\operatorname{Im}\widetilde{\Phi}_k$. The rest enters through Theorem A.

`Sanity.lean` takes countably many sectors, all equal to $\mathbb{C}$, so that $\bigoplus_k \mathcal{H}_k = \ell^2(\mathbb{N})$. For every $\varepsilon \ge 0$ the sectors $\mathbf{B}(\mathbb{C})$ are extended ε-C^* algebras whose block sum is the algebra of diagonal operators, which is infinite-dimensional (`ext_hyps_satisfiable`). The sectors $\Phi_k = \mathrm{id}$ satisfy every hypothesis of the channel statement for every $\eta \ge 0$, their block sum is the pinching $x \mapsto \bigoplus_k P_k x P_k$, and its $\widetilde{\Phi}$ has infinite rank (`channel_hyps_satisfiable`). So the channel statement is not a case of Theorem A, which needs $\widetilde{\Phi}$ of finite rank.

5.5 Step map

Step of the proof	Lean declarations (module)	Note
ℓ^∞ of C^* algebras is a C^* algebra	`lpCStarAlgebra`, `lpPartialOrder`, `lpStarOrderedRing` (LpCStar)	Mathlib gives only the non-unital structure; the unit needs every sector nonzero

Step of the proof	Lean declarations (module)	Note
matrix norms and positivity over ℓ^∞ are sectorwise	`norm_le_of_sectors`, `nonneg_of_sectors`, `lp_nonneg_iff` (LpCStar)	via the injective $*$-homomorphism $\mathbf{M}_n(\ell^\infty(\mathcal{A}_k)) \to \ell^\infty(\mathbf{M}_n(\mathcal{A}_k))$
$\mathbf{M}_n(\bigoplus^\infty \mathcal{A}_i) = \bigoplus^\infty \mathbf{M}_n(\mathcal{A}_i)$ isometrically	`bdHom` (HSum), `norm_le_of_BD`, `nonneg_of_BD` (BDMat)	the injective unital $*$-homomorphism $\ell^\infty(\mathbf{B}(\mathcal{H}_k)) \to \mathbf{B}(\bigoplus_k \mathcal{H}_k)$ is isometric at every matrix level
Proposition 3.1, extended: axioms pass to the sum	`extEps_sum` (Ext)	(2.1), (2.2), (2.5) sector by sector with the same ε
Proposition 3.1: $v = (v_i)_i$ is a δ-isomorphism	`iso_sum` (Ext)	surjectivity needs $\|v_i^{-1}\| \le (1-\delta)^{-1}$ uniformly
first sentence	`claimExt` (Proof)	sectors nonzero because $\|v_i(I) - I\| \le \delta < 1$; no sign hypothesis on `C₃`
Φ UCP, $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta$	`cp_of_sectors`, `cb_of_sectors`, `blk_cp`, `blk_cb` (Channel)	$x \mapsto P_k x P_k$ is the compression by the isometry $\mathcal{H}_k \to \bigoplus \mathcal{H}_k$
$\widetilde{\Phi}(X) = \bigoplus_k \widetilde{\Phi}_k(P_k X P_k)$	`dc_mul`, `dc_pow`, `dc_tsum`, `dc_tilde`, `summable_series` (Channel)	the series step needs the series for Φ itself to converge, so $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta$ comes first
$\widetilde{\Phi}$ takes block-diagonal values	`tilde_decomp`, `bdv_tilde` (Channel)	$\widetilde{\Phi} = \Phi + (\Phi - \frac{1}{2}) \sum_{j \ge 1} c_j N^j$ with $N = 4(\Phi - \Phi^2)$
$\operatorname{Im} \widetilde{\Phi} = \bigoplus_k^\infty \operatorname{Im} \widetilde{\Phi}_k$ with coordinatewise product	`blockSumAlg_tilde` (Channel)	the inclusion $\supseteq$ needs $\widetilde{\Phi}_k$ to fix $\operatorname{Im} \widetilde{\Phi}_k$ (`tilde_fix`, from Kitaev's Theorem 12.1)
$\Delta = \bigoplus_k \Delta_k$, $\Upsilon = (\Upsilon_k(P_k \cdot P_k))_k$ UCP	`deltaSum_ucp`, `upsSum_ucp` (ChannelSum)	
bounds as suprema	`channel_sum` (ChannelSum)	all bounds sector by sector with the same δ
constants	`claimChannel` (Proof)	$C = \|C_A\|$, $\eta_0 = \min(\eta_{0,A}, \eta_1, 1/8, 1/(2(\|C_A\| + 1)))$

5.6 What was not checked

The six assumptions; the first sentence for ε-C^* algebras in the sense of Definition 2.1 (the previous section); the ultraproduct remark after Theorem B and the open general case.

5.7 Findings

The proof is correct as written, and formalizing it made four of its quiet steps explicit.

The channel statement needs $C\eta < 1$. Proposition 3.1 needs the common δ to be less than 1, so that the inverse family is uniformly bounded. In the channel statement $\delta = C\eta$, and Theorem A's constants allow $C\eta_0$ to be as large as 5. The first version of the paper required only $\eta \le \eta_0$; it now also requires $C\eta < 1$. The formal version builds the condition into the threshold.

Identifying $\operatorname{Im} \widetilde{\Phi}$ uses Kitaev's Theorem 12.1 once more. Block-diagonality of $\widetilde{\Phi}$ gives $\operatorname{Im} \widetilde{\Phi} \subseteq \bigoplus_k^\infty \operatorname{Im} \widetilde{\Phi}_k$ at once. The reverse inclusion needs $\widetilde{\Phi}_k$ to fix $\operatorname{Im} \widetilde{\Phi}_k$, which follows from the exact unit of $(\operatorname{Im} \widetilde{\Phi}_k, \star)$.

The order of the argument matters. The termwise formula for $\widetilde{\Phi}$ needs the binomial series for Φ itself to converge, so the bound $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta$, a conclusion of the theorem, is proved first,

directly from the blocks.

The sector algebras are automatically nonzero. $\ell^\infty(\mathcal{B}_k)$ is a unital C^* algebra only if every $\mathcal{B}_k$ is nonzero, and this follows from $\|v_k(I) - I\| \le C\eta < 1$.

6 Further formal results

The library `LeanProof` in the same sources is an earlier project on the same toolchain. It builds cleanly and `#print axioms` reports only the standard axioms, but it was not run through the certificate audit. It contains `Kitaev.DeltaIso.symm` (the inverse of a δ-isomorphism is a $\delta/(1-\delta)^3$-isomorphism, Kitaev's "$\delta + O(\delta^2)$" with an explicit constant) and `Kitaev.Graded.graded_rigidity` (the paper's Proposition 4.3, graded rigidity of twisted group algebras).

7 Audit record and reproduction

Theorem A: PASS at 2026-10-16 11:48:58 +0800. Axioms of `main` and of all 8 non-vacuity theorems: `propext`, `Classical.choice`, `Quot.sound`. `leanchecker` passed on all 19 modules.

Theorem B, first sentence: PASS at 2026-10-16 11:49:20 +0800. Axioms of `main` and of all 9 non-vacuity theorems: `propext`, `Classical.choice`, `Quot.sound`. `leanchecker` passed on all 3 modules.

Theorem B, the rest: PASS at 2026-10-18 03:12:11 +0800. Axioms of `main` and of all 14 non-vacuity theorems: `propext`, `Classical.choice`, `Quot.sound`. `leanchecker` passed on all 10 modules.

The contents of `lean_sources.zip` were also built and audited as a standalone project, with the same result and the same hashes. To reproduce, with Lean installed through `elan`:

```
unzip lean_sources.zip
cd lean
lake exe cache get
lake build
lake env lean Check.lean
python3 audit.py KitaevChannels
python3 audit.py KitaevSectors
python3 audit.py KitaevSectorSums
```

Each audit ends with `PASS` and writes `Certs/<Name>/audit.json`, whose file hashes should match the tables below. Do not run `lake update`, since the proofs are checked against the pinned Mathlib commit.

`KitaevChannels` file	SHA-256
`Statement.lean`	`b6a7086255f94b4baf06ddc00f4c52b541bf4b68cada763c8dc85703f96417a5`
`Proof.lean`	`2b4708255626db0461cf13a08d696b92271e59f656ff5fda2957db833d3f46e7`
`Sanity.lean`	`eae5f53892b1b118ca11f4764c72fa33860e15c6b7ab7c66c464d7fa7280dc9f`
`Arith.lean`	`59a946bf3987db9df32d9d2efffa525b20ce1bac5daa1c3ea21947b7e5228bc2`
`Delta.lean`	`01cd54d71973ccb2bc2cc9c7ea10f93b95d5b1db2a94b1329991f7638c4c67df`
`DeltaProps.lean`	`76599f76831680efa15b3ddc7828eeb95f2a119f3523162818b847a4c6edc51c`
`DeltaUCP.lean`	`03850153cc55790cad1e61db15437e362df191b740ed4ce504429c75eb0c95e2`
`Design.lean`	`6b10cafa832b41a431aab1f0420cd4293692a6c88e6e45d8f02380e476c787e3`
`Final.lean`	`ee39e5a36c316270d7369df3e7cb55341faef0f65bb2e03538e49a027b18695e`
`Lemma124.lean`	`378f2162b7b8913035f6bcc367a3d196f475776b6c4c0e127e7774005f75cecb`
`Normalize.lean`	`1f5d1639df6a94c4b4a5fde1802e134fb2c1aff2f72026e4953018e719530d55`
`Objects.lean`	`e8bbabb7c1fff8341794871b16dae9982ffcc9e1ccfed53700cc6bd1a58e2552`
`Plumbing.lean`	`b9b0d08391c0345d67de87ce715bfb9ebbf9f673b129cbca325645129132c3ca`
`Stine.lean`	`3fc975542ce1d523623e9c0fe212f84d56176529ff39c005fe4a603fc4913027`
`Tilde.lean`	`f92f7e1873755fafb36f79f9d1ee0b6cce460bc146a01cf10bb79385f66556db`
`Toolkit.lean`	`5f3e38d7bb70cb4d90723931b36f6e26a55b8516cb2cace0d542e856e39e40f4`
`UpsBlock.lean`	`98323816af5ec84efccbfc2d3489b17f6bd87445b94ea2f1d4193d8af2734086`
`Upsilon.lean`	`4a5c64c5a5e3125858c26a8bd22a382d63d8258c88d24ed779e7a8a566afcb3f`

KitaevChannels file	SHA-256
Zeta.lean	caf5d2b286821a420c1baaa3e043d773e2eb3d1d5b487f363364241b444fee93
KitaevSectors file	**SHA-256**
Statement.lean	552f06db05a4dec26a6c25873796fc6034fba5e1848926262dfca810f34de59c
Proof.lean	1d36731b86abe6427a6e4238943de04e4bf697967389897312fe03bfb0d45090
Sanity.lean	e70e885cf940585ff174f39953731ca6d6391c32c5f8a7465dba05f85390d3a9
KitaevSectorSums file	**SHA-256**
Statement.lean	64072867c25a1ac44e071b2aee6ac6b5eb412a639acbe07ae7d31fa37f7b6d09
Proof.lean	bf212022058c858e45ccd0826e9e679c95d9e74fc82c0561f816ddfec5aa219f
Sanity.lean	e863b01ed18b72231a311901fe11de5400a50b4a8ba1f34573bc372d4690bd62
Assemble.lean	7f8a370444e9b5b3915d55286b1bd56b2a348feeb1efce0c44f5e2759a340b35
BDMat.lean	519dd09e433e02df39d566ad177460a1b9b67b93d1de170a2f177878181c93e5
Channel.lean	1c80e741eb8780b5d932c0b21325521a5557eb05e634c9f09a21b80aa561e73b
ChannelSum.lean	b5d8c70a4b6d7ef3379ff4e038b368b299f138b1f7b93f33b2f0eead3d705dae
Ext.lean	1f54b77867091a63db556e7533574ad10f0d9d06544b3ec83a558377a1e528a4
HSum.lean	938127ba2b67e81788b7d1aa1c1074b265ddb59d9f5637b50b182097a1543842
LpCStar.lean	d56500c5b81f0656aa8051ad47e20c6e7b901d487b287ce9bb479b3be1fb8967

A `Statement.lean` for Theorem A (verbatim)

```
import Mathlib.Analysis.CStarAlgebra.CStarMatrix
import Mathlib.Analysis.CStarAlgebra.ContinuousLinearMap
import Mathlib.Analysis.InnerProductSpace.StarOrder
import Mathlib.Algebra.Order.Star.Pi
import Mathlib.Analysis.Complex.Order
import Mathlib.Data.Nat.Choose.Central
import Mathlib.Topology.Algebra.InfiniteSum.Basic

/-!
# Theorem A: Kitaev's approximate factorization on an arbitrary Hilbert space (statement)

Source: `kitaev_almost_idempotent/note/infinite_dim.tex`, Theorem A, which extends
A. Kitaev, arXiv:2405.02434v2, Theorem 12.3 from `dim H < ∞` to Hilbert spaces of any dimension,
under the hypothesis that `Φ˜= θ(2Φ - 1)` has finite rank.

Six results are **assumed** (named hypotheses of `Claim`, never axioms): three textbook facts
about completely positive maps (`CPMapFacts`, `StinespringThm`, `UnitaryDesign`) and three results
of Kitaev's paper (Theorem 12.1, equation (12.6), Theorem 10.5). Everything else is proved.

`open scoped ComplexOrder` only switches on Mathlib's standard partial order on `ℂ`
(`z ≤ w ↔ re z ≤ re w ∧ im z = im w`), which Mathlib's matrix C*-norm requires.
-/

open scoped ComplexOrder

noncomputable section

namespace Cert.KitaevChannels

/-- `M_n(M)`: `n × n` matrices over a C*-algebra `M`, with Mathlib's C*-norm and order. -/
abbrev Mat (n : ℕ) (M : Type*) : Type _ := CStarMatrix (Fin n) (Fin n) M

/-- The finite-dimensional C*-algebra `⊕_{j<m} M_{d_j}(ℂ)`. -/
abbrev BlockAlg {m : ℕ} (d : Fin m → ℕ) : Type := (j : Fin m) → Mat (d j) ℂ

/-- `‖φ‖_cb ≤ c`: every amplification `1_{M_n} ⊗ φ` (apply `φ` entrywise to an `n × n`
matrix) has norm at most `c`. -/
def CBBound {M₁ M₂ : Type*} [CStarAlgebra M₁] [PartialOrder M₁] [StarOrderedRing M₁]
    [CStarAlgebra M₂] [PartialOrder M₂] [StarOrderedRing M₂] (φ : M₁ →ₗ[ℂ] M₂) (c : ℝ) : Prop :=
  ∀ (n : ℕ) (X : Mat n M₁), ‖X.map φ‖ ≤ c * ‖X‖

/-- `φ` is completely positive: every amplification maps positive matrices to positive matrices
```

```
(the defining condition of Mathlib's `CompletelyPositiveMap`). -/
def IsCP {M₁ M₂ : Type*} [CStarAlgebra M₁] [PartialOrder M₁] [StarOrderedRing M₁]
    [CStarAlgebra M₂] [PartialOrder M₂] [StarOrderedRing M₂] (φ : M₁ →ₗ[ℂ] M₂) : Prop :=
  ∀ (n : ℕ) (X : Mat n M₁), 0 ≤ X → 0 ≤ X.map φ

/-- `φ` is unital completely positive. -/
def IsUCP {M₁ M₂ : Type*} [CStarAlgebra M₁] [PartialOrder M₁] [StarOrderedRing M₁]
    [CStarAlgebra M₂] [PartialOrder M₂] [StarOrderedRing M₂] (φ : M₁ →ₗ[ℂ] M₂) : Prop :=
  φ 1 = 1 ∧ IsCP φ

/-- Kitaev (12.2): `Φ̃= θ(2Φ-1) = ½(1 + (2Φ-1)(1 - 4(Φ-Φ²))^{-1/2})`, with
`(1-x)^{-1/2} = ∑ₖ (2k choose k)/4ᵏ · xᵏ`, in the Banach algebra of bounded maps on `B(H)`. -/
def tildePhi {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
    (Φ : (H →L[ℂ] H) →L[ℂ] (H →L[ℂ] H)) : (H →L[ℂ] H) →L[ℂ] (H →L[ℂ] H) :=
  (1 / 2 : ℂ) • (1 + (2 • Φ - 1) *
    ∑' k : ℕ, ((Nat.centralBinom k : ℂ) / 4 ^ k) • (4 • (Φ - Φ * Φ)) ^ k)

/-- Kitaev (12.5): the approximate Choi–Effros product `X ⋆ Y = Φ̃(XY)`. -/
def choiEffros {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
    (Φ : (H →L[ℂ] H) →L[ℂ] (H →L[ℂ] H)) :
    (H →L[ℂ] H) →ₗ[ℂ] (H →L[ℂ] H) →ₗ[ℂ] (H →L[ℂ] H) :=
  (LinearMap.mul ℂ (H →L[ℂ] H)).compr₂ (tildePhi Φ).toLinearMap

/-- The matrix product over a bilinear (possibly non-associative) product `mul`:
`(X ⋆ Y)ᵢⱼ = ∑ₗ mul (Xᵢₗ) (Yₗⱼ)` (Kitaev §10). -/
def matMul {M : Type*} [AddCommMonoid M] [Module ℂ M] (mul : M →ₗ[ℂ] M →ₗ[ℂ] M) {n : ℕ}
    (X Y : Mat n M) : Mat n M :=
  CStarMatrix.ofMatrix (fun i j => ∑ l, mul (X i l) (Y l j))

/-- Kitaev's extended ε-C* algebra (§10), realized concretely inside `B(H)`, with exact unit:
a subspace `A ∋ 1` closed under `†` and under the bilinear product `mul`, with
`(X ⋆ Y)† = Y† ⋆ X†` and `X ⋆ 1 = X = 1 ⋆ X`, such that for every `n` the space `M_n(A)`, with
the norm of `M_n(B(H))`, the entrywise involution and the matrix product over `mul`, satisfies
(2.1), (2.2) and (2.5). (The remaining axioms (2.4), (2.6) hold automatically in `M_n(B(H))`.) -/
def ExtEpsCStarSub {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
    (ε : ℝ) (A : Submodule ℂ (H →L[ℂ] H))
    (mul : (H →L[ℂ] H) →ₗ[ℂ] (H →L[ℂ] H) →ₗ[ℂ] (H →L[ℂ] H)) : Prop :=
  (1 : H →L[ℂ] H) ∈ A ∧ (∀ x ∈ A, star x ∈ A) ∧ (∀ x ∈ A, ∀ y ∈ A, mul x y ∈ A) ∧
  (∀ x ∈ A, ∀ y ∈ A, star (mul x y) = mul (star y) (star x)) ∧
  (∀ x ∈ A, mul x 1 = x ∧ mul 1 x = x) ∧
  ∀ (n : ℕ) (X Y Z : Mat n (H →L[ℂ] H)),
    (∀ i j, X i j ∈ A) → (∀ i j, Y i j ∈ A) → (∀ i j, Z i j ∈ A) →
    ‖matMul mul X Y‖ ≤ (1 + ε) * ‖X‖ * ‖Y‖ ∧
    ‖matMul mul (matMul mul X Y) Z - matMul mul X (matMul mul Y Z)‖ ≤ ε * ‖X‖ * ‖Y‖ * ‖Z‖ ∧
    (1 - ε) * ‖X‖ ^ 2 ≤ ‖matMul mul (star X) X‖

/-- Kitaev's extended δ-isomorphism (§10) from `⊕ⱼ M_{dⱼ}(ℂ)` onto `(A, mul)`: a linear bijection
onto `A` commuting with `†`, with `‖v(I) - I‖ ≤ δ`, and at every matrix level `n`
`‖v(XY) - v(X) ⋆ v(Y)‖ ≤ δ‖X‖‖Y‖` and `(1-δ)‖X‖ ≤ ‖v(X)‖ ≤ (1+δ)‖X‖`. -/
def IsExtDeltaIsoOnto {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
    {m : ℕ} {d : Fin m → ℕ} (δ : ℝ) (A : Submodule ℂ (H →L[ℂ] H))
    (mul : (H →L[ℂ] H) →ₗ[ℂ] (H →L[ℂ] H) →ₗ[ℂ] (H →L[ℂ] H))
    (v : BlockAlg d →ₗ[ℂ] (H →L[ℂ] H)) : Prop :=
  Function.Injective v ∧ LinearMap.range v = A ∧
  (∀ b, v (star b) = star (v b)) ∧ ‖v 1 - 1‖ ≤ δ ∧
  ∀ (n : ℕ) (X Y : Mat n (BlockAlg d)),
    ‖(X * Y).map v - matMul mul (X.map v) (Y.map v)‖ ≤ δ * ‖X‖ * ‖Y‖ ∧
    (1 - δ) * ‖X‖ ≤ ‖X.map v‖ ∧ ‖X.map v‖ ≤ (1 + δ) * ‖X‖

/-- **Assumed (textbook): completely positive maps.** For a CP map `φ` between unital
C*-algebras, `‖φ‖_cb = ‖φ(1)‖` (Paulsen, *Completely bounded maps and operator algebras*,
Prop. 3.6), and `φ` commutes with the involution (positive maps are self-adjoint). -/
def CPMapFacts : Prop :=
  ∀ (M₁ M₂ : Type) [CStarAlgebra M₁] [PartialOrder M₁] [StarOrderedRing M₁]
    [CStarAlgebra M₂] [PartialOrder M₂] [StarOrderedRing M₂] (φ : M₁ →ₗ[ℂ] M₂),
    IsCP φ → CBBound φ ‖φ 1‖ ∧ ∀ x, φ (star x) = star (φ x)

/-- **Assumed (textbook): Stinespring's theorem** (Stinespring 1955; Paulsen Thm. 4.1 and the
```

```
easy converse): a linear map `φ : M → B(H)` from a unital C*-algebra is completely positive iff
`φ(x) = V† π(x) V` for a unital *-representation `π` of `M` on some Hilbert space `K` and a
bounded `V : H → K`. -/
def StinespringThm : Prop :=
  ∀ (M : Type) [CStarAlgebra M] [PartialOrder M] [StarOrderedRing M]
    (H : Type) [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
    (φ : M →ₗ[ℂ] (H →L[ℂ] H)),
    IsCP φ ↔ ∃ (K : Type) (_ : NormedAddCommGroup K) (_ : InnerProductSpace ℂ K)
      (_ : CompleteSpace K) (π : M →⋆ₐ[ℂ] (K →L[ℂ] K)) (V : H →L[ℂ] K),
      ∀ x, φ x = (ContinuousLinearMap.adjoint V) ∘L (π x) ∘L V

/-- **Assumed (textbook): unitary 1-designs** (e.g. the generalized Pauli group): for every `d`
there is a finite probability distribution `p` on unitaries `U s` of `M_d(ℂ)` with the twirl
identity `∑ₛ pₛ (Uₛ†)ₐᵦ (Uₛ)_{c e} = δₐₑ δ_{b c} / d`. -/
def UnitaryDesign : Prop :=
  ∀ d : ℕ, ∃ (S : Type) (_ : Fintype S) (p : S → ℝ) (U : S → Mat d ℂ),
    (∀ s, 0 ≤ p s) ∧ ∑ s, p s = 1 ∧ (∀ s, star (U s) * U s = 1 ∧ U s * star (U s) = 1) ∧
    ∀ a b c e : Fin d, ∑ s, (p s : ℂ) * star (U s b a) * U s c e =
      if a = e ∧ b = c then 1 / (d : ℂ) else 0

/-- **Assumed: Kitaev, Theorem 12.1** (proved there for Hilbert spaces of any dimension, §12.1):
for an η-idempotent UCP map, `Im Φ̃` with the product `Φ̃(XY)` is an extended `O(η)`-C* algebra. -/
def Kitaev12_1 (c₁ η₁ : ℝ) : Prop :=
  ∀ (H : Type) [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H] [Nontrivial H]
    (Φ : (H →L[ℂ] H) →L[ℂ] (H →L[ℂ] H)) (η : ℝ),
    IsUCP Φ.toLinearMap → CBBound (Φ * Φ - Φ).toLinearMap η → 0 ≤ η → η < η₁ →
    ExtEpsCStarSub (c₁ * η) (LinearMap.range (tildePhi Φ).toLinearMap) (choiEffros Φ)

/-- **Assumed: Kitaev, equation (12.6)** (proved there for any dimension, applied to `1_{M_n} ⊗ Φ`):
`Φ(Φ(Φ(X)Φ(Y))Φ(Z)) = Φ(Φ(X)Φ(Y)Φ(Z)) + O(η)‖X‖‖Y‖‖Z‖`. -/
def Kitaev12_6 (c₂ η₂ : ℝ) : Prop :=
  ∀ (H : Type) [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H] [Nontrivial H]
    (Φ : (H →L[ℂ] H) →L[ℂ] (H →L[ℂ] H)) (η : ℝ),
    IsUCP Φ.toLinearMap → CBBound (Φ * Φ - Φ).toLinearMap η → 0 ≤ η → η < η₂ →
    ∀ (n : ℕ) (X Y Z : Mat n (H →L[ℂ] H)),
      ‖(((X.map Φ * Y.map Φ).map Φ) * Z.map Φ).map Φ - (X.map Φ * Y.map Φ * Z.map Φ).map Φ‖
        ≤ c₂ * η * ‖X‖ * ‖Y‖ * ‖Z‖

/-- **Assumed: Kitaev, Theorem 10.5**, for extended ε-C* algebras realized inside some `B(H)`,
with the C*-algebra in the block form `⊕ⱼ M_{dⱼ}(ℂ)` that Kitaev's proof constructs (§9). -/
def Kitaev10_5 (C₃ ε₃ : ℝ) : Prop :=
  ∀ (H : Type) [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H] [Nontrivial H]
    (A : Submodule ℂ (H →L[ℂ] H)) [FiniteDimensional ℂ A]
    (mul : (H →L[ℂ] H) →ₗ[ℂ] (H →L[ℂ] H) →ₗ[ℂ] (H →L[ℂ] H)) (ε : ℝ),
    0 ≤ ε → ε < ε₃ → ExtEpsCStarSub ε A mul →
    ∃ (m : ℕ) (d : Fin m → ℕ) (v : BlockAlg d →ₗ[ℂ] (H →L[ℂ] H)),
      IsExtDeltaIsoOnto (C₃ * ε) A mul v

/-- **Theorem A.** Assume the three textbook facts above, and Kitaev's Theorem 12.1, (12.6) and
Theorem 10.5 (with their constants).
Then there are universal constants `η₀ > 0` and `C` such that for every nonzero Hilbert space `H`
of any dimension and every UCP map `Φ` on `B(H)` with `‖Φ² - Φ‖_cb ≤ η < η₀` and `Φ̃` of finite
rank, there are a block algebra `B = ⊕ⱼ M_{dⱼ}(ℂ)` and
(i) an extended `(C η)`-isomorphism `v : B → (Im Φ̃, ⋆)`, in particular a linear bijection, and
(ii) UCP maps `Δ : B → B(H)`, `Y : B(H) → B` with `‖ΔY - Φ‖_cb ≤ C η` and
`‖Yₙ(Δₙ(X) Δₙ(Y)) - XY‖ ≤ C η ‖X‖‖Y‖` for all `n` and `X, Y ∈ Mₙ(B)`. -/
def Claim : Prop :=
  CPMapFacts → StinespringThm → UnitaryDesign →
  ∀ (c₁ η₁ c₂ η₂ C₃ ε₃ : ℝ), 0 ≤ c₁ → 0 < η₁ → 0 ≤ c₂ → 0 < η₂ → 0 ≤ C₃ → 0 < ε₃ →
  Kitaev12_1 c₁ η₁ → Kitaev12_6 c₂ η₂ → Kitaev10_5 C₃ ε₃ →
  ∃ (η₀ C : ℝ), 0 < η₀ ∧
  ∀ (H : Type) [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H] [Nontrivial H]
    (Φ : (H →L[ℂ] H) →L[ℂ] (H →L[ℂ] H)) (η : ℝ),
    IsUCP Φ.toLinearMap → CBBound (Φ * Φ - Φ).toLinearMap η → 0 ≤ η → η < η₀ →
    FiniteDimensional ℂ (LinearMap.range (tildePhi Φ).toLinearMap) →
    ∃ (m : ℕ) (d : Fin m → ℕ) (v : BlockAlg d →ₗ[ℂ] (H →L[ℂ] H))
      (Δ : BlockAlg d →ₗ[ℂ] (H →L[ℂ] H)) (Y : (H →L[ℂ] H) →ₗ[ℂ] BlockAlg d),
      IsExtDeltaIsoOnto (C * η) (LinearMap.range (tildePhi Φ).toLinearMap) (choiEffros Φ) v ∧
```

```
      IsUCP Δ ∧ IsUCP Y ∧
      CBBound (Δ ∘ₗ Y - Φ.toLinearMap) (C * η) ∧
      ∀ (n : ℕ) (X Y : Mat n (BlockAlg d)),
        ‖(X.map Δ * Y.map Δ).map Y - X * Y‖ ≤ C * η * ‖X‖ * ‖Y‖

end Cert.KitaevChannels
```

B `Statement.lean` for Theorem B, first sentence (verbatim)

```
import Mathlib.Analysis.CStarAlgebra.Basic
import Mathlib.Analysis.CStarAlgebra.Classes
import Mathlib.Analysis.Normed.Lp.lpSpace
import Mathlib.LinearAlgebra.FiniteDimensional.Defs

/-!
# Kitaev's rigidity theorem for infinite-dimensional sector sums (statement)

Source: A. Kitaev, *Almost-idempotent quantum channels and approximate C*-algebras*,
arXiv:2405.02434v2, Definitions 2.1–2.2 and Theorem 2.3; and the companion note
`kitaev_almost_idempotent/note/infinite_dim.tex`, Proposition 3.1 / Theorem B.

Everything Kitaev defines is transcribed literally below. Kitaev's Theorem 2.3 (finite dimension)
is **not** proved here: it appears as the named hypothesis `KitaevFiniteDim C ε₀` of `Claim`.
-/

open scoped ComplexConjugate ENNReal

noncomputable section

namespace Cert.KitaevSectors

/-- Kitaev's Definition 2.1: an ε-C* algebra structure on a complex normed space `A`, with the
default approximate unit (2.6). The product is bilinear but **not** assumed associative. -/
structure EpsCStar (ε : ℝ) (A : Type*) [NormedAddCommGroup A] [NormedSpace ℂ A] where
  /-- the bilinear product `XY` -/
  mul : A → A → A
  mul_add : ∀ x y z, mul x (y + z) = mul x y + mul x z
  add_mul : ∀ x y z, mul (x + y) z = mul x z + mul y z
  mul_smul : ∀ (c : ℂ) (x y : A), mul x (c • y) = c • mul x y
  smul_mul : ∀ (c : ℂ) (x y : A), mul (c • x) y = c • mul x y
  /-- the involution `X ↦ X†` (conjugate linear) -/
  star : A → A
  star_add : ∀ x y, star (x + y) = star x + star y
  star_smul : ∀ (c : ℂ) (x : A), star (c • x) = conj c • star x
  star_star : ∀ x, star (star x) = x
  /-- (2.4) `‖X†‖ = ‖X‖` -/
  norm_star : ∀ x, ‖star x‖ = ‖x‖
  /-- (2.4) `(XY)† = Y†X†` -/
  star_mul : ∀ x y, star (mul x y) = mul (star y) (star x)
  /-- the unit `I`, with `I† = I` -/
  one : A
  star_one : star one = one
  /-- (2.1) `‖XY‖ ≤ (1+ε)‖X‖‖Y‖` -/
  norm_mul_le : ∀ x y, ‖mul x y‖ ≤ (1 + ε) * ‖x‖ * ‖y‖
  /-- (2.2) `‖(XY)Z − X(YZ)‖ ≤ ε‖X‖‖Y‖‖Z‖` -/
  assoc_le : ∀ x y z, ‖mul (mul x y) z - mul x (mul y z)‖ ≤ ε * ‖x‖ * ‖y‖ * ‖z‖
  /-- (2.5) `‖X†X‖ ≥ (1−ε)‖X‖²` -/
  cstar_le : ∀ x, (1 - ε) * ‖x‖ ^ 2 ≤ ‖mul (star x) x‖
  /-- (2.6) `‖XI − X‖ ≤ ε‖X‖` -/
  mul_one_le : ∀ x, ‖mul x one - x‖ ≤ ε * ‖x‖
  /-- (2.6) `‖IX − X‖ ≤ ε‖X‖` -/
  one_mul_le : ∀ x, ‖mul one x - x‖ ≤ ε * ‖x‖
  /-- (2.6) `|‖I‖ − 1| ≤ ε` -/
  norm_one_le : |‖one‖ - 1| ≤ ε
```

```
/-- Kitaev's Definition 2.2: a δ-isomorphism is a linear bijection commuting with the involution,
with (2.8) `‖v(I) − I‖ ≤ δ`, (2.9) `‖v(XY) − v(X)v(Y)‖ ≤ δ‖X‖‖Y‖`, and the δ-inclusion bounds
`(1−δ)‖X‖ ≤ ‖v(X)‖ ≤ (1+δ)‖X‖`. -/
structure DeltaIso (δ : ℝ) {A A' : Type*} [NormedAddCommGroup A] [NormedSpace ℂ A]
    [NormedAddCommGroup A'] [NormedSpace ℂ A'] {ε ε' : ℝ}
    (S : EpsCStar ε A) (S' : EpsCStar ε' A') where
  /-- the linear bijection `v` -/
  toEquiv : A ≃ₗ[ℂ] A'
  map_star : ∀ x, toEquiv (S.star x) = S'.star (toEquiv x)
  map_one_le : ‖toEquiv S.one - S'.one‖ ≤ δ
  map_mul_le : ∀ x y, ‖toEquiv (S.mul x y) - S'.mul (toEquiv x) (toEquiv y)‖ ≤ δ * ‖x‖ * ‖y‖
  le_norm_map : ∀ x, (1 - δ) * ‖x‖ ≤ ‖toEquiv x‖
  norm_map_le : ∀ x, ‖toEquiv x‖ ≤ (1 + δ) * ‖x‖

/-- A genuine (unital, nonzero) C*-algebra, regarded as a 0-C* algebra: its own product,
involution and unit. -/
def ofCStarAlgebra (B : Type*) [CStarAlgebra B] [Nontrivial B] : EpsCStar 0 B where
  mul x y := x * y
  mul_add := mul_add
  add_mul := add_mul
  mul_smul c x y := mul_smul_comm c x y
  smul_mul c x y := smul_mul_assoc c x y
  star := star
  star_add := star_add
  star_smul c x := star_smul c x
  star_star := star_star
  norm_star := norm_star
  star_mul := star_mul
  one := 1
  star_one := star_one B
  norm_mul_le x y := by simpa using norm_mul_le x y
  assoc_le x y z := by simp [mul_assoc]
  cstar_le x := by simp [CStarRing.norm_star_mul_self, sq]
  mul_one_le x := by simp
  one_mul_le x := by simp
  norm_one_le := by simp

/-- **Kitaev's Theorem 2.3 (assumed, not proved here)**, with its constants made explicit:
every finite-dimensional ε-C* algebra with `0 ≤ ε < ε₀` is `(C ε)`-isomorphic to some
(nonzero, unital) C*-algebra `B`. -/
def KitaevFiniteDim (C ε₀ : ℝ) : Prop :=
  ∀ (A : Type) [NormedAddCommGroup A] [NormedSpace ℂ A] [CompleteSpace A]
    [FiniteDimensional ℂ A] (ε : ℝ) (S : EpsCStar ε A), 0 ≤ ε → ε < ε₀ →
    ∃ (B : Type) (_ : CStarAlgebra B) (_ : Nontrivial B),
      Nonempty (DeltaIso (C * ε) (ofCStarAlgebra B) S)

/-- **Theorem B (infinitely many finite sectors).** If Kitaev's Theorem 2.3 holds with
constants `C, ε₀`, then for every `0 ≤ ε < ε₀` with `C ε < 1`, every nonempty family of
finite-dimensional ε-C* algebras `A i`, and every ε-C* structure `T` on the ℓ^∞-direct sum
`lp A ∞` (bounded families, sup norm) whose product, involution and unit are computed
coordinatewise, there is a C*-algebra `B` and a `(C ε)`-isomorphism `B → (lp A ∞, T)` —
in particular a linear bijection. The direct sum is infinite-dimensional whenever infinitely
many `A i` are nonzero. -/
def Claim : Prop :=
  ∀ (C ε₀ : ℝ), KitaevFiniteDim C ε₀ →
  ∀ (ε : ℝ), 0 ≤ ε → ε < ε₀ → C * ε < 1 →
  ∀ (ι : Type) [Nonempty ι] (A : ι → Type) [∀ i, NormedAddCommGroup (A i)]
    [∀ i, NormedSpace ℂ (A i)] [∀ i, FiniteDimensional ℂ (A i)]
    (S : ∀ i, EpsCStar ε (A i)) (T : EpsCStar ε (lp A ∞)),
    (∀ f g i, T.mul f g i = (S i).mul (f i) (g i)) →
    (∀ f i, T.star f i = (S i).star (f i)) →
    (∀ i, T.one i = (S i).one) →
    ∃ (B : Type) (_ : CStarAlgebra B) (_ : Nontrivial B),
      Nonempty (DeltaIso (C * ε) (ofCStarAlgebra B) T)

end Cert.KitaevSectors
```

C `Statement.lean` for Theorem B, the rest (verbatim)

```
import Certs.KitaevChannels.Statement
import Mathlib.Analysis.InnerProductSpace.l2Space

/-!
# Theorem B, the parts not covered by `KitaevSectors` (statement)

Source: the paper *Kitaev's ε-C\* rigidity beyond finite dimensions* (J. Naskar and Claude),
Theorem B. The certificate `KitaevSectors` proves its first sentence for ε-C\* algebras in the sense
of Kitaev's Definition 2.1. This certificate proves the rest.

* `ClaimExt` is the first sentence for *extended* ε-C\* algebras (Kitaev §10). As in the
  certificate of Theorem A, an extended ε-C\* algebra is realized as a subspace `Aₖ ⊆ B(Hₖ)` with a
  bilinear product `mulₖ`, with the matrix norms of `Mₙ(B(Hₖ))`. The ℓ^∞-direct sum is realized
  on the Hilbert direct sum `⊕ₖ Hₖ` (`lp H 2`): it consists of the block-diagonal operators whose
  `k`-th diagonal block lies in `Aₖ`, with the blockwise product. Assumes Kitaev's Theorem 10.5.
* `ClaimChannel` is the second sentence. For a block-diagonal channel `Φ = ⊕ₖ Φₖ(Pₖ · Pₖ)` on
  `B(⊕ₖ Hₖ)` whose sectors satisfy the hypotheses of Theorem A, `Φ` is UCP with
  `‖Φ² - Φ‖_cb ≤ η`, and the conclusions of Theorem A hold with a C\* algebra in place of the
  finite-dimensional block algebra. Assumes what Theorem A assumes.

Kitaev's results enter only as the named hypotheses `Kitaev12_1`, `Kitaev12_6`, `Kitaev10_5`
(defined in `Certs.KitaevChannels.Statement`), never as axioms.
-/

open scoped ComplexOrder ENNReal

noncomputable section

namespace Cert.KitaevSectorSums

open Cert.KitaevChannels

section Blocks

variable {ι : Type} [DecidableEq ι] {H : ι → Type} [∀ k, NormedAddCommGroup (H k)]
  [∀ k, InnerProductSpace ℂ (H k)]

/-- The inclusion `Hₖ → ⊕ₖ Hₖ` of the `k`-th summand of the Hilbert direct sum `lp H 2`. -/
def inj (k : ι) : H k →L[ℂ] lp H 2 := lp.singleContinuousLinearMap ℂ H 2 k

/-- The projection `⊕ₖ Hₖ → Hₖ` onto the `k`-th summand. -/
def prj (k : ι) : lp H 2 →L[ℂ] H k := lp.evalCLM ℂ H 2 k

/-- The `(k, l)` block `Pₖ x Pₗ` of an operator `x` on `⊕ₖ Hₖ`, as an operator `Hₗ → Hₖ`. -/
def blk (k l : ι) (x : lp H 2 →L[ℂ] lp H 2) : H l →L[ℂ] H k := prj k ∘L x ∘L inj l

/-- `Φ` is the block-diagonal channel `x ↦ ⊕ₖ Φₖ(Pₖ x Pₖ)`: every off-diagonal block of `Φ(x)`
vanishes, and the `k`-th diagonal block of `Φ(x)` is `Φₖ` applied to the `k`-th diagonal block
of `x`. -/
def IsBlockSumMap (Φ : (lp H 2 →L[ℂ] lp H 2) →L[ℂ] (lp H 2 →L[ℂ] lp H 2))
    (Φk : ∀ k, (H k →L[ℂ] H k) →L[ℂ] (H k →L[ℂ] H k)) : Prop :=
  ∀ x, (∀ k, blk k k (Φ x) = Φk k (blk k k x)) ∧ ∀ k l, k ≠ l → blk k l (Φ x) = 0

/-- `(A, mul)` is the ℓ^∞-direct sum of the `(Aₖ, mulₖ)`, realized block-diagonally: `A` is the
set of bounded operators on `⊕ₖ Hₖ` with vanishing off-diagonal blocks and `k`-th diagonal block
in `Aₖ`, and for `x, y ∈ A` the product `mul x y` is block-diagonal with `k`-th block
`mulₖ xₖ yₖ`. -/
def IsBlockSumAlg (A : Submodule ℂ (lp H 2 →L[ℂ] lp H 2))
    (mul : (lp H 2 →L[ℂ] lp H 2) →ₗ[ℂ] (lp H 2 →L[ℂ] lp H 2) →ₗ[ℂ] (lp H 2 →L[ℂ] lp H 2))
    (Ak : ∀ k, Submodule ℂ (H k →L[ℂ] H k))
    (mulk : ∀ k, (H k →L[ℂ] H k) →ₗ[ℂ] (H k →L[ℂ] H k) →ₗ[ℂ] (H k →L[ℂ] H k)) : Prop :=
  (∀ x, x ∈ A ↔ (∀ k l, k ≠ l → blk k l x = 0) ∧ ∀ k, blk k k x ∈ Ak k) ∧
  ∀ x ∈ A, ∀ y ∈ A, (∀ k, blk k k (mul x y) = mulk k (blk k k x) (blk k k y)) ∧
    ∀ k l, k ≠ l → blk k l (mul x y) = 0
```

```
end Blocks

/-- Kitaev's extended δ-isomorphism (§10) from an arbitrary C\* algebra `B` onto `(A, mul)`:
the same conditions as `IsExtDeltaIsoOnto`, whose domain is a finite block algebra. -/
def IsExtDeltaIsoOntoGen {B : Type} [CStarAlgebra B] [PartialOrder B] [StarOrderedRing B]
    {K : Type} [NormedAddCommGroup K] [InnerProductSpace ℂ K] [CompleteSpace K]
    (δ : ℝ) (A : Submodule ℂ (K →L[ℂ] K)) (mul : (K →L[ℂ] K) →ₗ[ℂ] (K →L[ℂ] K) →ₗ[ℂ] (K →L[ℂ] K))
    (v : B →ₗ[ℂ] (K →L[ℂ] K)) : Prop :=
  Function.Injective v ∧ LinearMap.range v = A ∧
  (∀ b, v (star b) = star (v b)) ∧ ‖v 1 - 1‖ ≤ δ ∧
  ∀ (n : ℕ) (X Y : Mat n B),
    ‖(X * Y).map v - matMul mul (X.map v) (Y.map v)‖ ≤ δ * ‖X‖ * ‖Y‖ ∧
    (1 - δ) * ‖X‖ ≤ ‖X.map v‖ ∧ ‖X.map v‖ ≤ (1 + δ) * ‖X‖

/-- **Theorem B, first sentence, extended version.** If Kitaev's Theorem 10.5 holds with
constants `C₃, ε₃`, then for `0 ≤ ε < ε₃` with `C₃ ε < 1`, the ℓ^∞-direct sum of any nonempty
family of finite-dimensional extended ε-C\* algebras is an extended ε-C\* algebra and is extended
`(C₃ ε)`-isomorphic to a C\* algebra. -/
def ClaimExt : Prop :=
  ∀ (C₃ ε₃ : ℝ), Kitaev10_5 C₃ ε₃ →
  ∀ (ε : ℝ), 0 ≤ ε → ε < ε₃ → C₃ * ε < 1 →
  ∀ (ι : Type) [DecidableEq ι] [Nonempty ι] (H : ι → Type) [∀ k, NormedAddCommGroup (H k)]
    [∀ k, InnerProductSpace ℂ (H k)] [∀ k, CompleteSpace (H k)] [∀ k, Nontrivial (H k)]
    (Ak : ∀ k, Submodule ℂ (H k →L[ℂ] H k)) [∀ k, FiniteDimensional ℂ (Ak k)]
    (mulk : ∀ k, (H k →L[ℂ] H k) →ₗ[ℂ] (H k →L[ℂ] H k) →ₗ[ℂ] (H k →L[ℂ] H k))
    (A : Submodule ℂ (lp H 2 →L[ℂ] lp H 2))
    (mul : (lp H 2 →L[ℂ] lp H 2) →ₗ[ℂ] (lp H 2 →L[ℂ] lp H 2) →ₗ[ℂ] (lp H 2 →L[ℂ] lp H 2)),
    (∀ k, ExtEpsCStarSub ε (Ak k) (mulk k)) → IsBlockSumAlg A mul Ak mulk →
    ExtEpsCStarSub ε A mul ∧
    ∃ (B : Type) (_ : CStarAlgebra B) (_ : PartialOrder B) (_ : StarOrderedRing B)
      (v : B →ₗ[ℂ] (lp H 2 →L[ℂ] lp H 2)), IsExtDeltaIsoOntoGen (C₃ * ε) A mul v

/-- **Theorem B, second sentence.** Assume what Theorem A assumes. There are constants
`η₀ > 0` and `C` such that for every nonempty family of nonzero Hilbert spaces `Hₖ` and UCP maps
`Φₖ` on `B(Hₖ)` with `‖Φₖ² - Φₖ‖_cb ≤ η < η₀` and `Φ̃ₖ` of finite rank, the block-diagonal map
`Φ = ⊕ₖ Φₖ(Pₖ · Pₖ)` on `B(⊕ₖ Hₖ)` is UCP with `‖Φ² - Φ‖_cb ≤ η`, and there are a C\* algebra
`B`, an extended `C η`-isomorphism `v : B → (Im Φ̃, ⋆)`, and UCP maps `Δ : B → B(⊕ₖ Hₖ)`,
`Y : B(⊕ₖ Hₖ) → B` with `‖ΔY - Φ‖_cb ≤ C η` and `‖Yₙ(Δₙ(X) Δₙ(Y)) - XY‖ ≤ C η ‖X‖‖Y‖`. -/
def ClaimChannel : Prop :=
  CPMapFacts → StinespringThm → UnitaryDesign →
  ∀ (c₁ η₁ c₂ η₂ C₃ ε₃ : ℝ), 0 ≤ c₁ → 0 < η₁ → 0 ≤ c₂ → 0 < η₂ → 0 ≤ C₃ → 0 < ε₃ →
  Kitaev12_1 c₁ η₁ → Kitaev12_6 c₂ η₂ → Kitaev10_5 C₃ ε₃ →
  ∃ (η₀ C : ℝ), 0 < η₀ ∧
  ∀ (ι : Type) [DecidableEq ι] [Nonempty ι] (H : ι → Type) [∀ k, NormedAddCommGroup (H k)]
    [∀ k, InnerProductSpace ℂ (H k)] [∀ k, CompleteSpace (H k)] [∀ k, Nontrivial (H k)]
    (Φk : ∀ k, (H k →L[ℂ] H k) →L[ℂ] (H k →L[ℂ] H k))
    (Φ : (lp H 2 →L[ℂ] lp H 2) →L[ℂ] (lp H 2 →L[ℂ] lp H 2)) (η : ℝ),
    (∀ k, IsUCP (Φk k).toLinearMap) → (∀ k, CBBound (Φk k * Φk k - Φk k).toLinearMap η) →
    0 ≤ η → η < η₀ →
    (∀ k, FiniteDimensional ℂ (LinearMap.range (tildePhi (Φk k)).toLinearMap)) →
    IsBlockSumMap Φ Φk →
    IsUCP Φ.toLinearMap ∧ CBBound (Φ * Φ - Φ).toLinearMap η ∧
    ∃ (B : Type) (_ : CStarAlgebra B) (_ : PartialOrder B) (_ : StarOrderedRing B)
      (v Δ : B →ₗ[ℂ] (lp H 2 →L[ℂ] lp H 2)) (Y : (lp H 2 →L[ℂ] lp H 2) →ₗ[ℂ] B),
      IsExtDeltaIsoOntoGen (C * η) (LinearMap.range (tildePhi Φ).toLinearMap) (choiEffros Φ) v ∧
      IsUCP Δ ∧ IsUCP Y ∧
      CBBound (Δ ∘ₗ Y - Φ.toLinearMap) (C * η) ∧
      ∀ (n : ℕ) (X Y : Mat n B),
        ‖(X.map Δ * Y.map Δ).map Y - X * Y‖ ≤ C * η * ‖X‖ * ‖Y‖

/-- The rest of Theorem B: its first sentence for extended ε-C\* algebras, and its second
sentence (block-diagonal channels). -/
def Claim : Prop := ClaimExt ∧ ClaimChannel

end Cert.KitaevSectorSums
```